Chemical Equilibrium and Kc

Answer key with worked solutions

A. Understanding

1. Dynamic equilibrium.

> Both the forward and reverse reactions continue, at **equal rates**, so the concentrations of reactants and products stay **constant**. They are generally **not equal**.

B. Writing Kc

2. Expressions.

> (a) $\frac{[NH_3]^2}{[N_2][H_2]^3}$ (b) $\frac{[SO_3]^2}{[SO_2]^2[O_2]}$ (c) $[CO_2]$ (solids left out) (d) $\frac{[PCl_3][Cl_2]}{[PCl_5]}$

C. Calculating Kc

3. Kc for SO3.

> $$K_c = \frac{(0.160\ \text{M})^2}{(0.0400\ \text{M})^2(0.0200\ \text{M})} \times 1\ \text{M} = \mathbf{800.}$$
>
> Dividing each concentration by $c^\circ = 1$ M makes the units cancel: $\frac{0.0256}{(0.00160)(0.0200)} = 800.$

4. Related reactions.

> (a) Reverse: $\frac{1}{800.} = \mathbf{1.25 \times 10^{-3}}$ (b) Coefficients halved: $\sqrt{800.} = \mathbf{28.3}$

D. Q versus K

5. Q for N2O4.

> $Q = \frac{(0.0200\ \text{M})^2}{0.100\ \text{M}} \times \frac{1}{1\ \text{M}} = \frac{0.000400}{0.100} = \mathbf{0.00400}$. $Q < K_c$ (0.0200), so the reaction goes **forward**: more NO_2 forms.

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E. ICE tables

6. H_2 + I_2 starting at 0.200 M each.

> Let x be the decrease in $[H_2]$, in M. Equilibrium: $[H_2] = [I_2] = 0.200\ \text{M} - x$; $[HI] = 2x$.
> $\sqrt{49.0} = 7.00 = \dfrac{2x}{0.200\ \text{M} - x}$, so $1.400\ \text{M} - 7.00x = 2x$ and $x = \dfrac{1.400\ \text{M}}{9.00} = 0.1556\ \text{M}$
> $[HI] = \mathbf{0.311\ M}$; $[H_2] = [I_2] = 0.200\ \text{M} - 0.1556\ \text{M} = \mathbf{0.0444\ M}$

7. Kc for PCl_5.

> The Cl_2 formed (0.100 M) equals the PCl_3 formed and the PCl_5 used up. Equilibrium: $[PCl_5] = 0.500\ \text{M} - 0.100\ \text{M} = 0.400\ \text{M}$; $[PCl_3] = [Cl_2] = 0.100\ \text{M}$.
> $K_c = \dfrac{(0.100\ \text{M})(0.100\ \text{M})}{0.400\ \text{M}} \times \dfrac{1}{1\ \text{M}} = \mathbf{0.0250}$

F. Challenge

8. Large and small K.

> $K = 1 \times 10^{10}$: the mixture is almost entirely **products** (the reaction goes essentially to completion). $K = 1 \times 10^{-10}$: almost entirely **reactants** (hardly any reaction).

9. PCl_5 starting at 0.300 M.

> Let x (in M) = $[PCl_3] = [Cl_2]$ at equilibrium; $[PCl_5] = 0.300\ \text{M} - x$. Working in units of M (so each concentration divided by 1 M):
>
> $$0.0250 = \frac{x^2}{0.300 - x} \;\Rightarrow\; x^2 + 0.0250x - 0.00750 = 0$$
>
> $$x = \frac{-0.0250 + \sqrt{(0.0250)^2 + 4(0.00750)}}{2} = \frac{-0.0250 + 0.1750}{2} = 0.0750$$
>
> $[PCl_3] = [Cl_2] = \mathbf{0.0750\ M}$; $[PCl_5] = 0.300\ \text{M} - 0.0750\ \text{M} = \mathbf{0.225\ M}$. (The other root is negative, so it is rejected.)

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